DEPARTMENT OF MATHEMATICS
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

CLASS TEST-I

ENGINEERING MATHEMATICS – IV

(CSE Branch) Probability & Statistics

Class: B.E
Semester: IV
Academic Year: 2024-2025

Max Marks: 30
Duration: 1 hr30 min
Date:27 -03-2025

		Marks	BT	CO
	PART A **Answer All Questions (10 M)**			
1. a)	Find the median of first 7 composite numbers.	2	L5	1
b)	Find the expected value of the following distribution : x: -10, -20, 30, 75, 80 P(X): 1/5, 3/20, 1/2, 1/10, 1/20	2	L3	1
c)	Find the Mean value in Gamma densities	2	L2	2
d)	Find the mean of the Discrete Probability distribution is given by $f(x)=\frac{1}{n}$ *for* $x=1,2,3,.....n$	2	L5	1
e)	For an Exponential distribution $\lambda = 1.2$ *find* (*i*) $P(X \geq 0.5)$ (*ii*) $P(1 \leq X \leq 2)$ Also find the Mean and Variance.	2	L6	2
	PART B **Answer All Questions (20 M)**			
2. a)	The density function of a random variable X is $f(x) = \{ e^{-x}, \ x \geq 0$; 0, *otherwise*. Then Find E(X), E(X^2) and Var (X) .	5	L2	1
b)	A random variable X is defined as the sum of the numbers of the faces when two dice are thrown. Find the mean of X.	5	L3	1
3. a)	The values of $\mu_1, \mu_2, \mu_3,$ *and* μ_4 are 0, 9.2, 3.6, and 1.22 respectively. Find out the skewness and Kurtosis of the distribution	5	L2	1
b)	The following table shows the ages of the Patients admitted in a hospital on a particular day. Age(in years): 5-15, 15-25, 25-35, 35-45, 45-55, 55-65 No. of Patients: 6, 11, 21, 23, 14, 5	5	L2	1

Find the Mode

4 a)	Find the Mean and Variance in Exponential Distribution	5	L1	
b)	If a random variable X has the gamma distribution with $r = 2$ and $\lambda = \frac{1}{2}$ fins the Mean and S.D and the probability that X will take a value less than 4.	5	L2	2

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